Basic Mechanics

-ME 162-

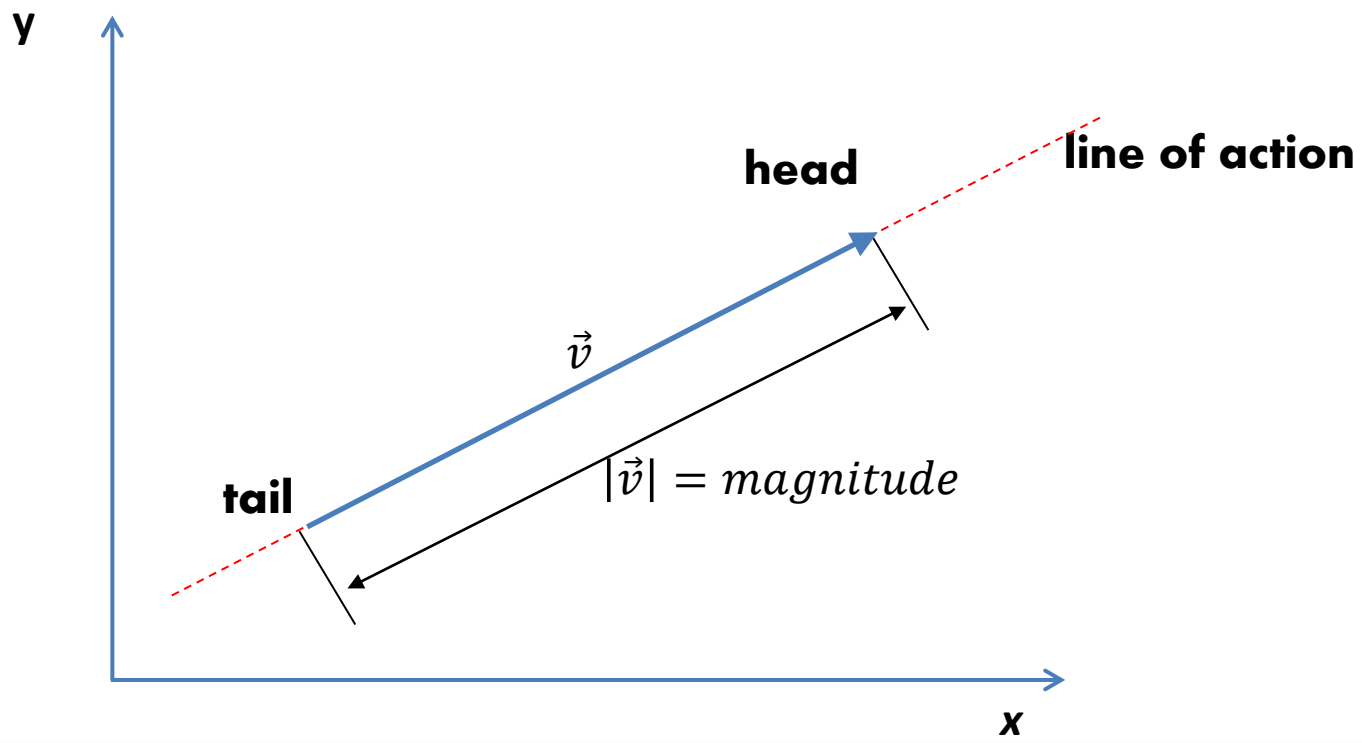
Vector Representation of Forces

Department of Mechanical Engineering, KNUST

D. A. Quansah, PhD

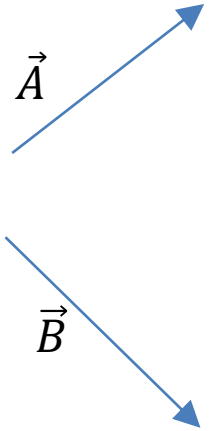
Vectors and representation of forces

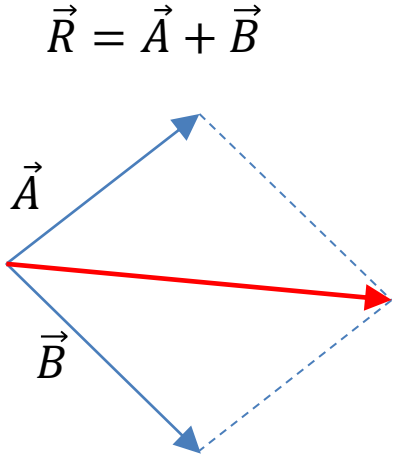
Vector nomenclature:



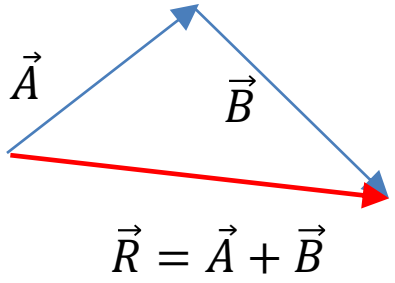
Vectors and representation of forces

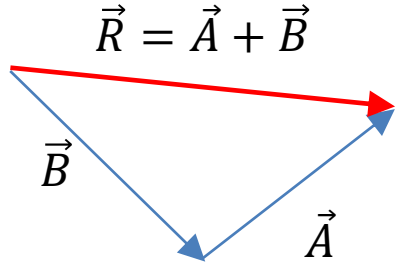
Basic vector operations:





tail-to-tail addition





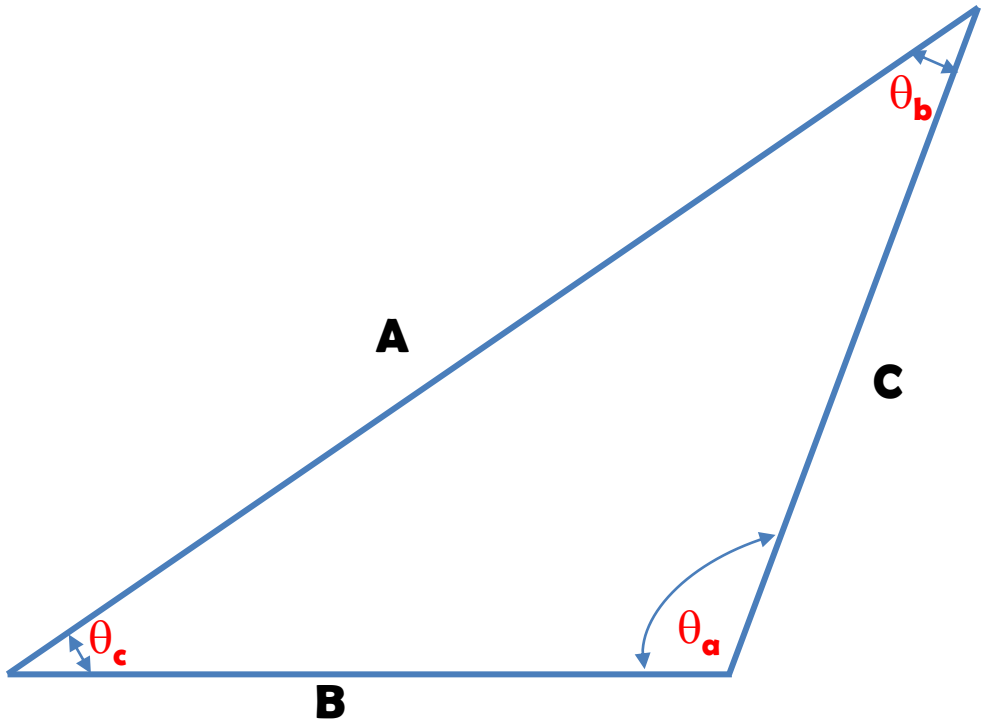
head-to-tail addition

- **Two vectors are equivalent if they have same magnitude and orientation.**
- **Equivalent vectors may have different lines of action, provided they are parallel.**

Vectors and representation of forces

$\vec{A} + \vec{B} = \vec{B} + \vec{A}$	Vector addition is cummutative
$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$	Associative property of vector addition
$\vec{R} = s\vec{A} = \vec{A}s$	Multiplication of a vector by a scalar
$s(\vec{A} + \vec{B}) = s\vec{B} + s\vec{A}$	Distributive property of vector addition
$(s + t)\vec{A} = s\vec{A} + t\vec{A}$	Distributive property – addition of scalars
$(st)\vec{A} = s(t\vec{A})$	Associative property – multiplication by a scalar

Law of Sines and Cosines



$$\frac{\sin\theta_a}{A} = \frac{\sin\theta_b}{B} = \frac{\sin\theta_c}{C}$$

Law of sines

Law of cosines

$$A = \sqrt{B^2 + C^2 - 2BC\cos\theta_a}$$

$$B = \sqrt{A^2 + C^2 - 2AC\cos\theta_b}$$

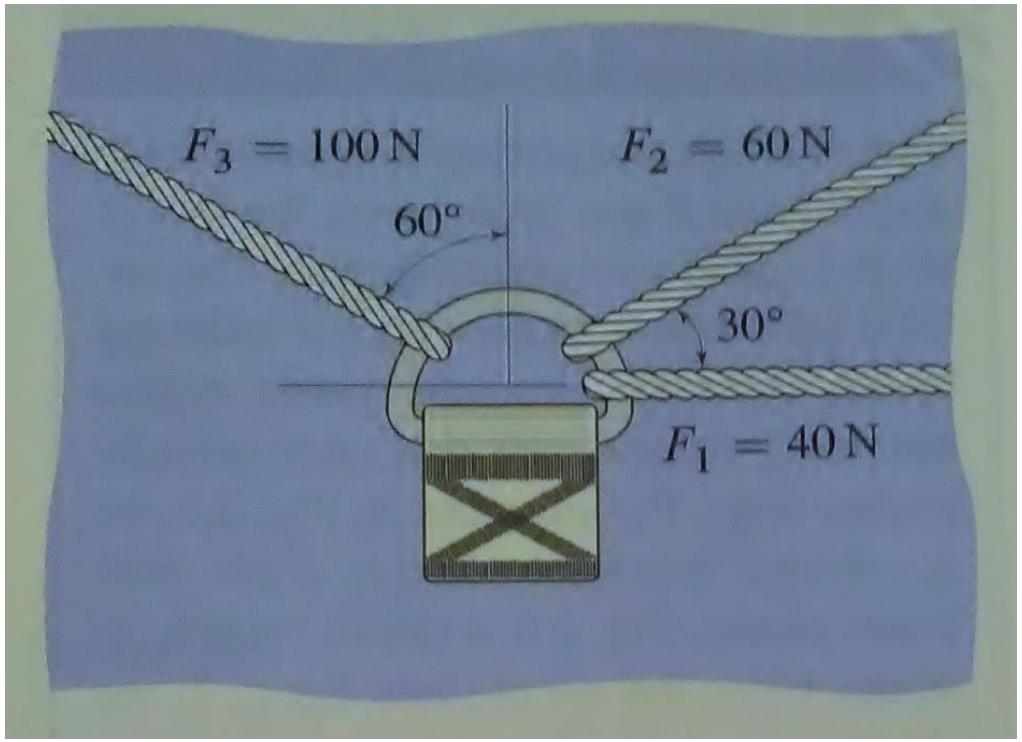
$$C = \sqrt{A^2 + B^2 - 2AB\cos\theta_c}$$

Application

Example:

A D-ring is sewn on a backpack for use in securing miscellaneous items to the outside of the backpack.

If the D-ring has three cords tied to it and support forces shown in the Figure beside, determine the resultant force applied to the D ring by the cords, expressing the result as a vector.



Application

Key notes:

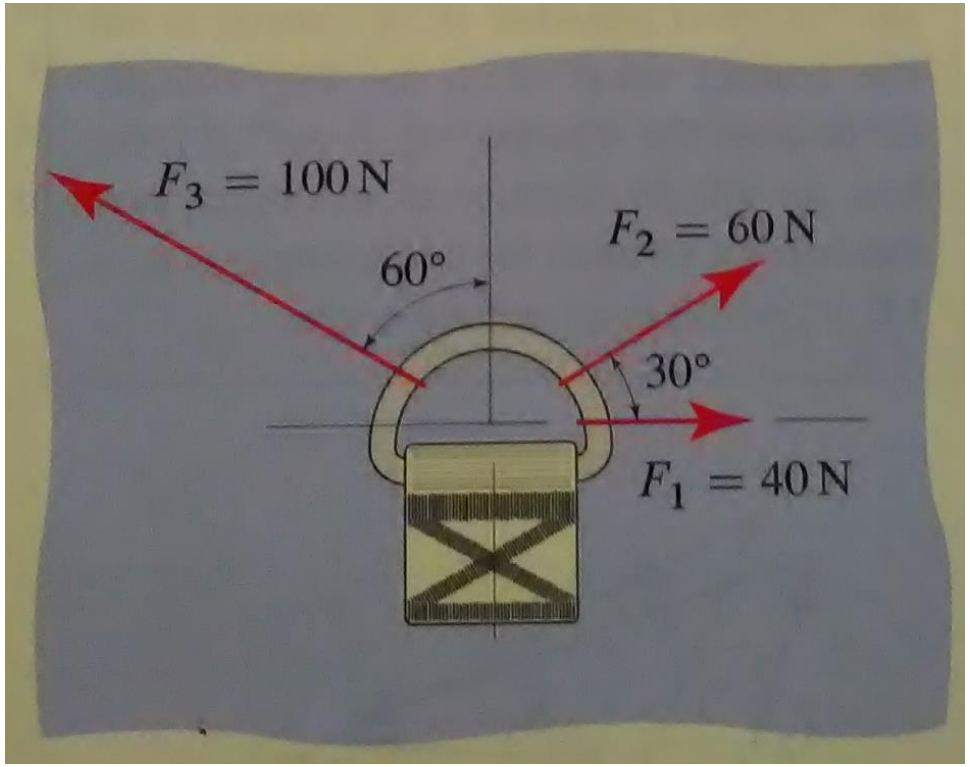
A force supported by a cord has the same direction as the cord.

Figure can therefore be redrawn as shown.

The resultant vector is the sum of all 3 force vectors.

Add the force vectors using head-to-tail method.

Apply the law of sines and cosines to determine the results of the vector addition.

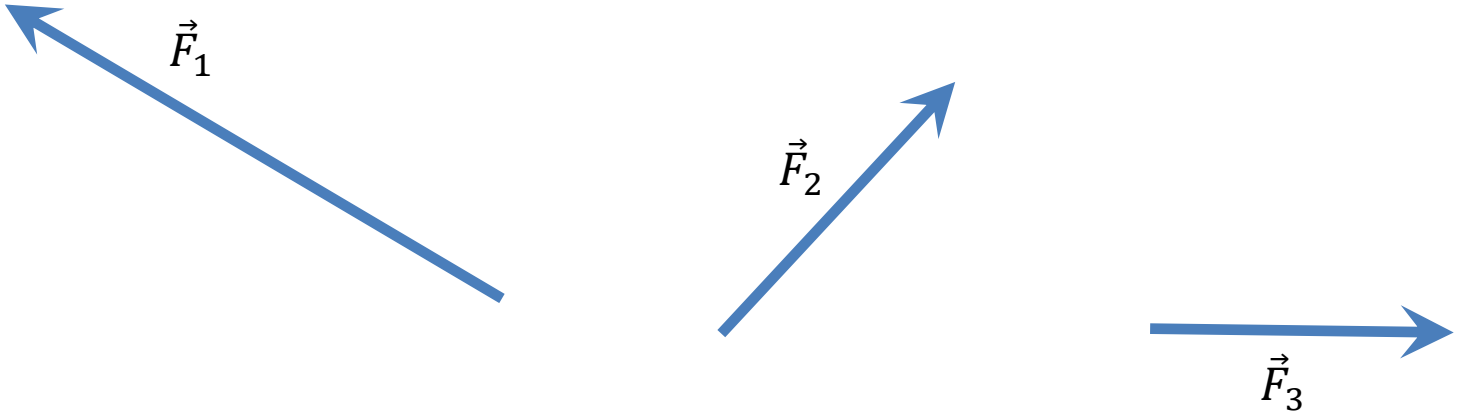


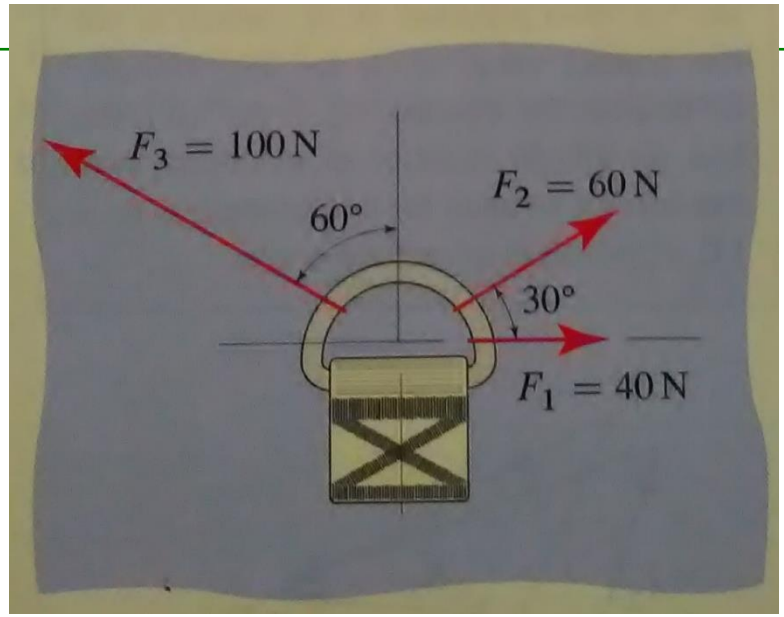
Application

Key notes:

The Resultant for may be written as:

$$\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

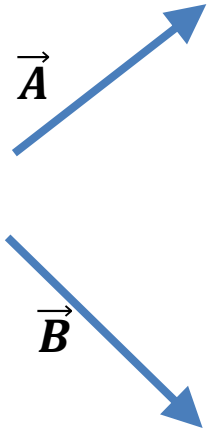


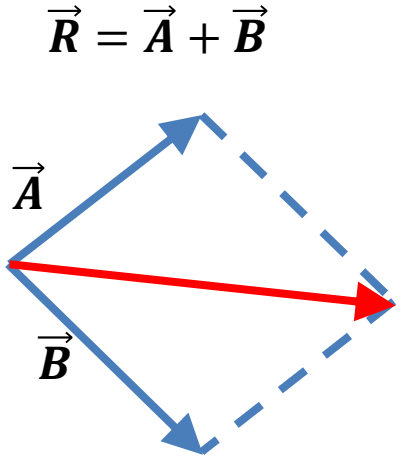


Before we continue, recall the following:

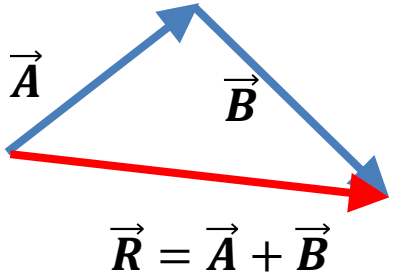
Vectors and representation of forces

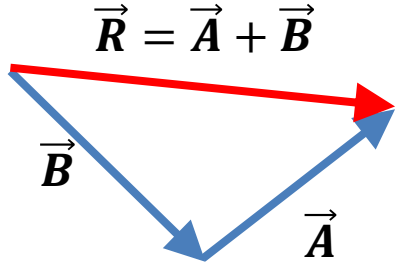
Basic vector operations:





tail-to-tail addition (parallelogram addition)



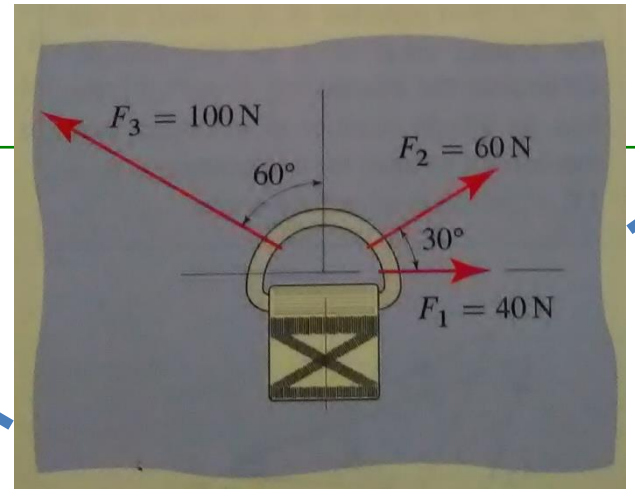


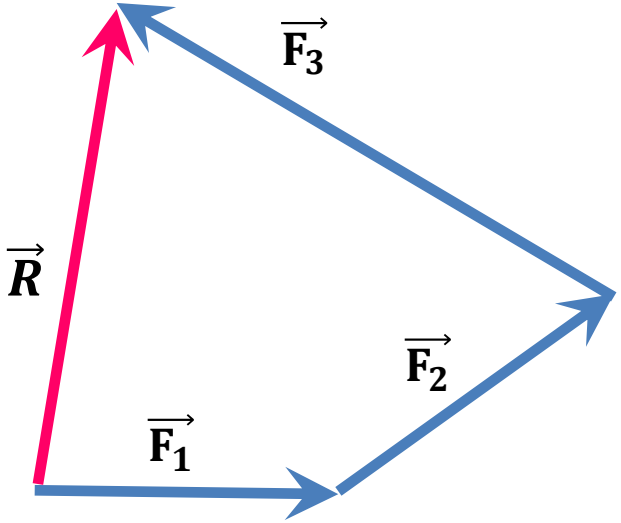
head-to-tail addition

- **Two vectors are equivalent if they have same magnitude and orientation.**
- **Equivalent vectors may have different lines of action, provided they are parallel.**

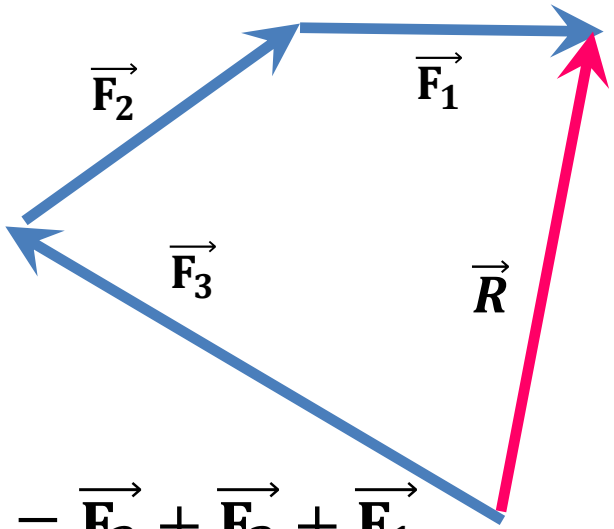
Application

The system of forces may be represented by force polygons as follows:

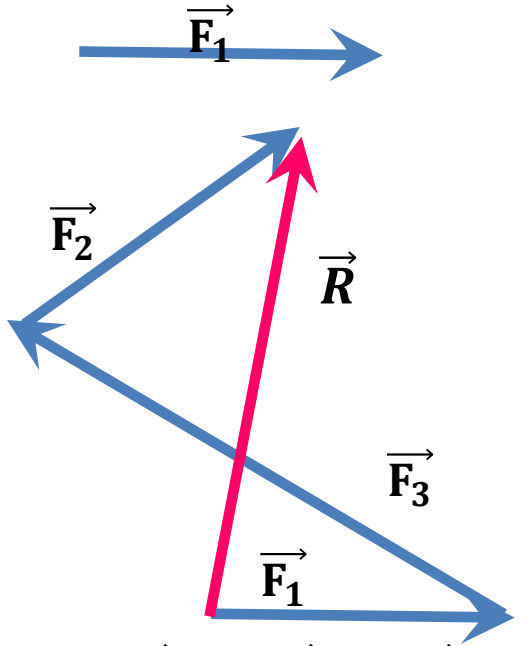




$$\vec{R} = \vec{F_1} + \vec{F_2} + \vec{F_3}$$



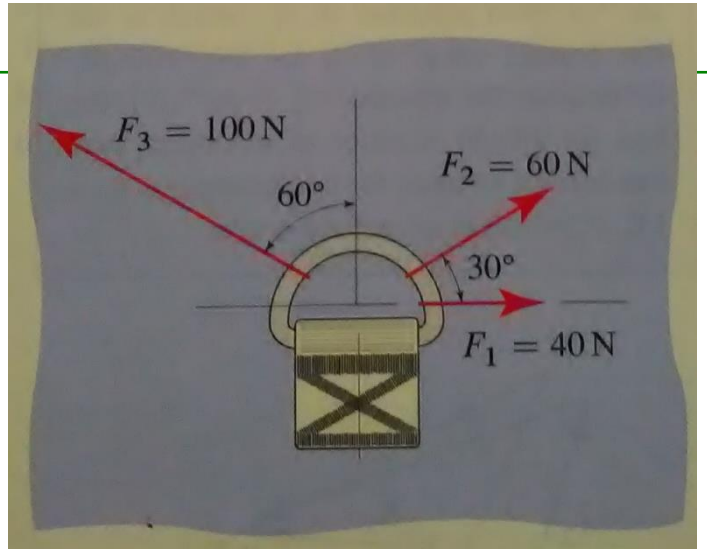
$$\vec{R} = \vec{F_3} + \vec{F_2} + \vec{F_1}$$



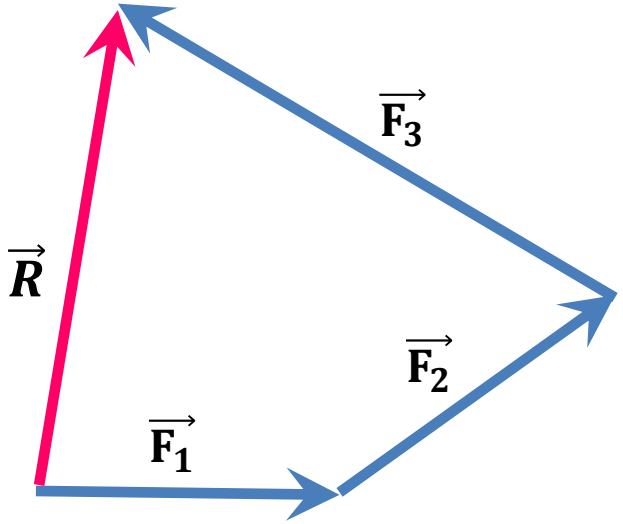
$$\vec{R} = \vec{F_1} + \vec{F_3} + \vec{F_2}$$

Recall: vector addition is commutative.

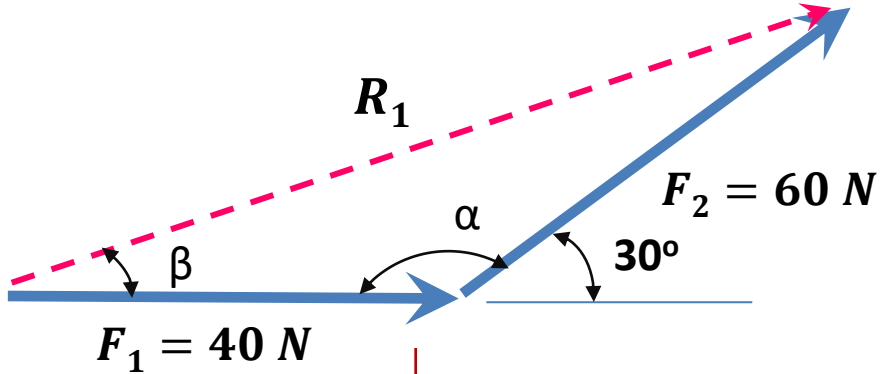
Application



Simpler geometries may be created from the force polygons for easier analysis:



$$\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

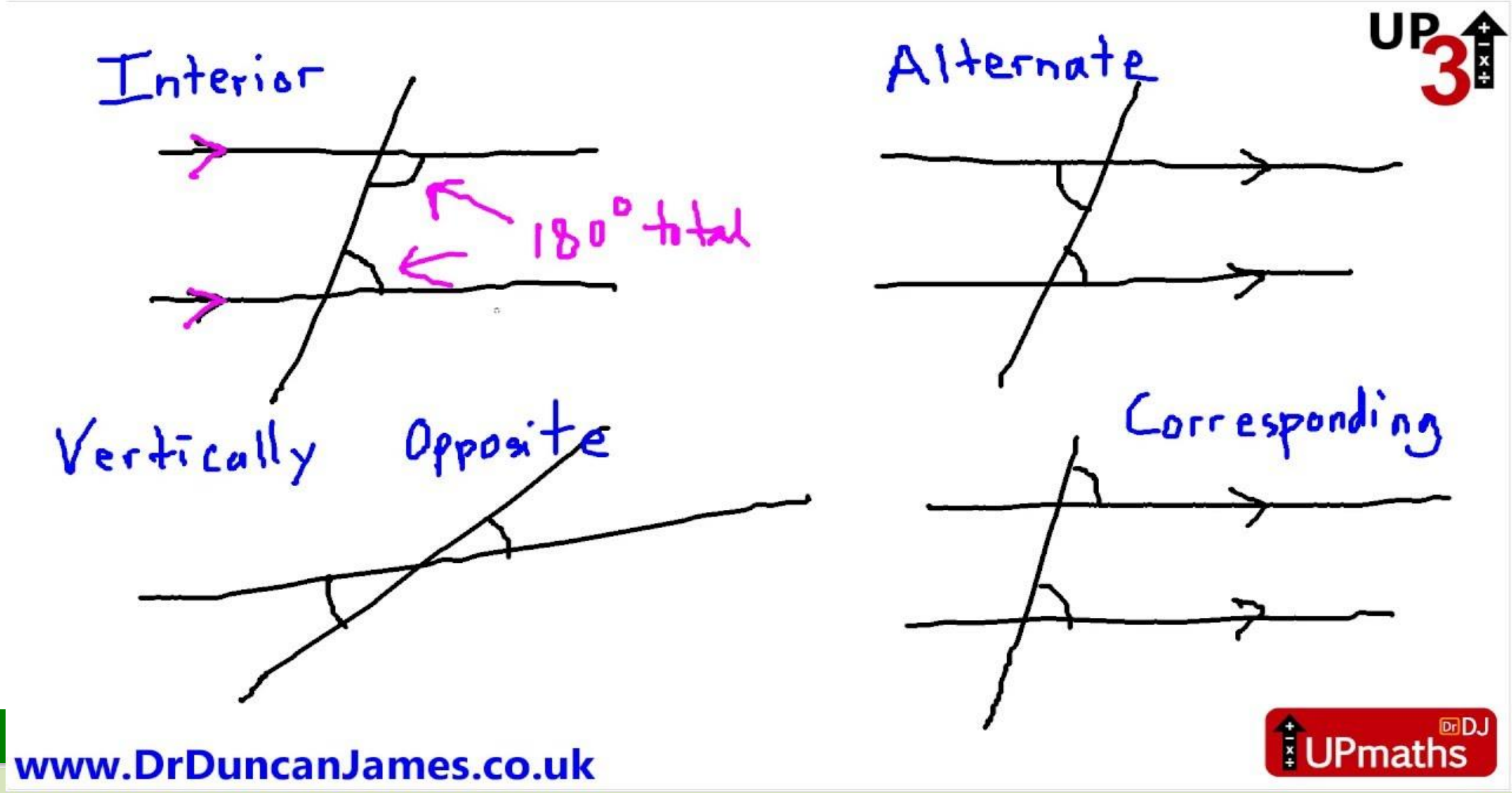


$$R_1 = \sqrt{F_1^2 + F_2^2 - 2F_1F_2\cos\alpha}$$

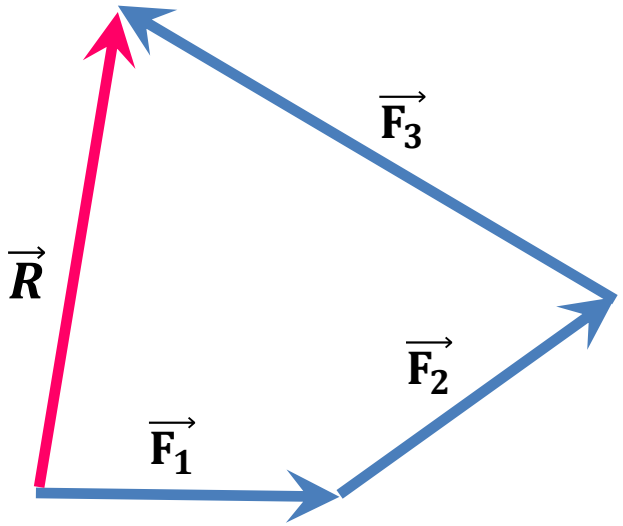
$$R_1 = \sqrt{(40\ N)^2 + (60\ N)^2 - 2(40\ N)(60\ N)(\cos 150^o)} = 96.73\ N$$

$$\alpha = 180^o - 30^o$$

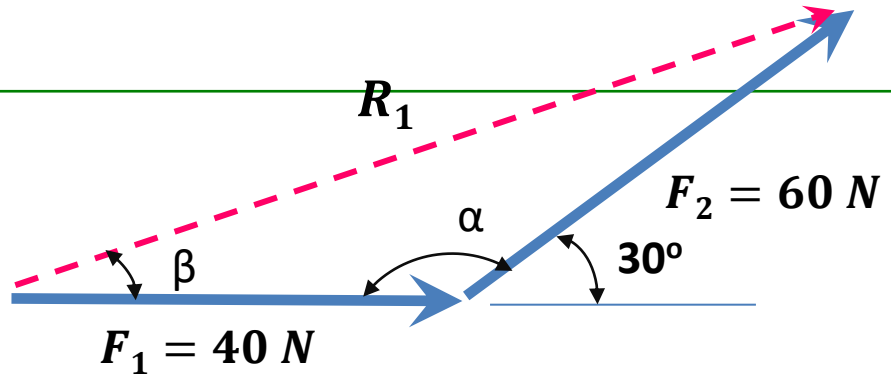
On Angles - A Refresher



Application



$$\vec{R} = \vec{F_1} + \vec{F_2} + \vec{F_3}$$



$$R_1 = 96.73\ N$$

The law of sines is then applied to determine angle β:

$$\frac{\sin\alpha}{R_1} = \frac{\sin\beta}{F_2}$$

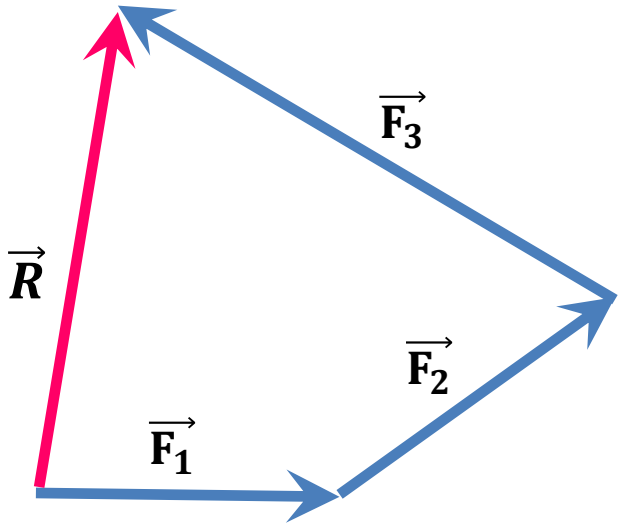
$$\sin\beta = F_2 \frac{\sin\alpha}{R_1}$$

$$\sin\beta = \frac{60\ \text{N}}{96.73\ \text{N}} \sin(150^o)$$

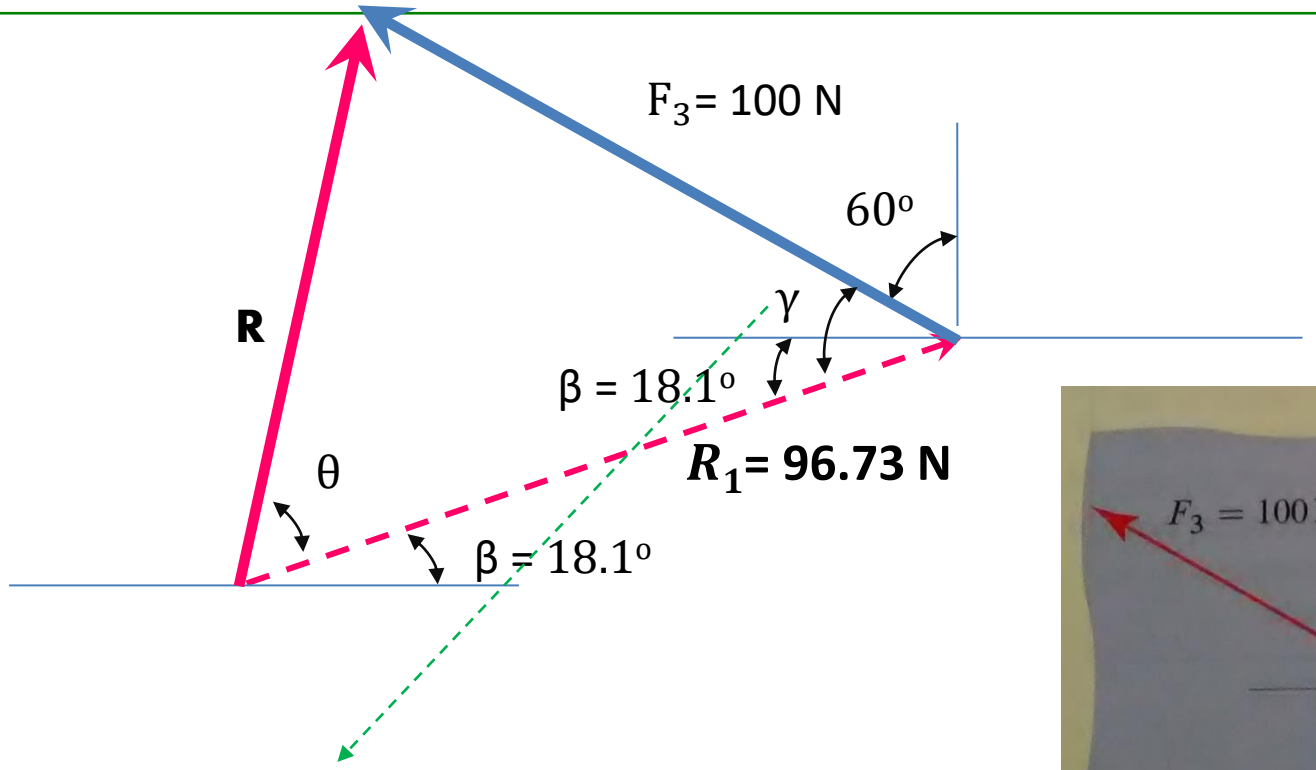
$$= 0.3101$$

$$\beta = \sin^{-1}(0.3101) = 18.1^o$$

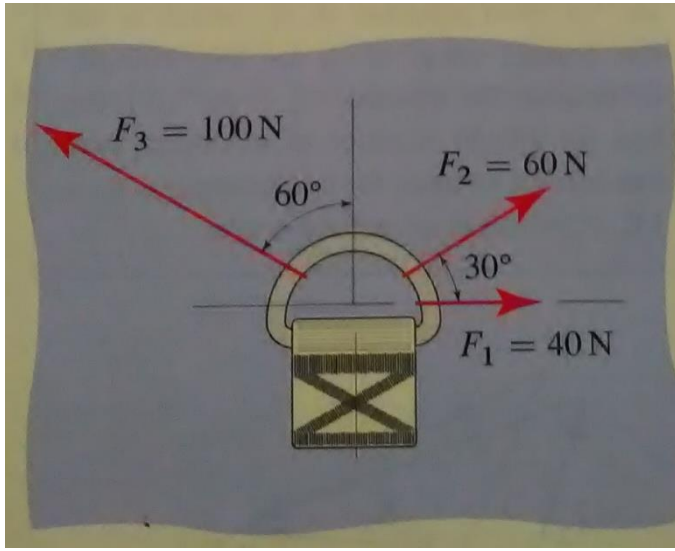
Application



$$\vec{R} = \vec{F_1} + \vec{F_2} + \vec{F_3}$$



$$\gamma = 90^o - 60^o + 18.1^o = 48.1^o$$

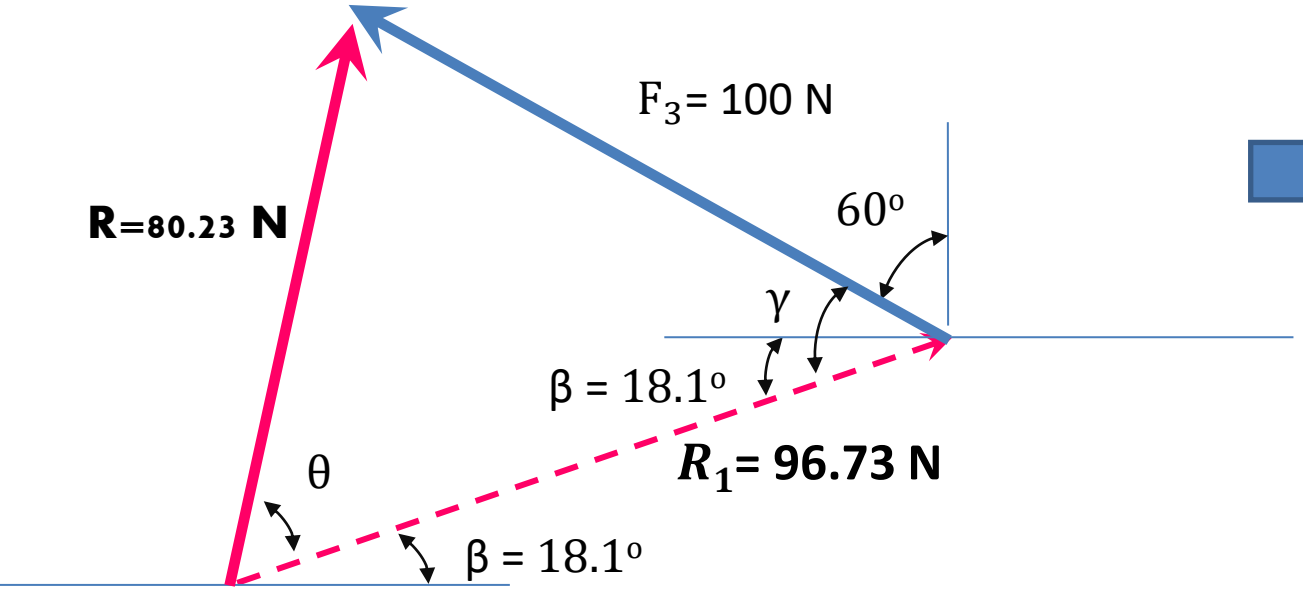


$$R = \sqrt{R_1^2 + F_3^2 - 2R_1F_3\cos\gamma}$$

$$R = \sqrt{96.73^2 + 100^2 - 2(96.73)(100)\cos(48.1^o)}$$

$$= 80.23 \text{ N}$$

Application



$$\frac{F_3}{\sin\theta} = \frac{R}{\sin\gamma}$$

$$\sin\theta = \frac{F_3}{R}\sin\gamma$$

$$\sin\theta = \frac{100\ N}{80.2\ N}\sin(48.1^o) = 0.928$$

$$\theta = \sin^{-1}(0.928) = 68.1^o$$

$$\vec{R} = 80.2\ \text{N}\ @\ 86.2^o$$

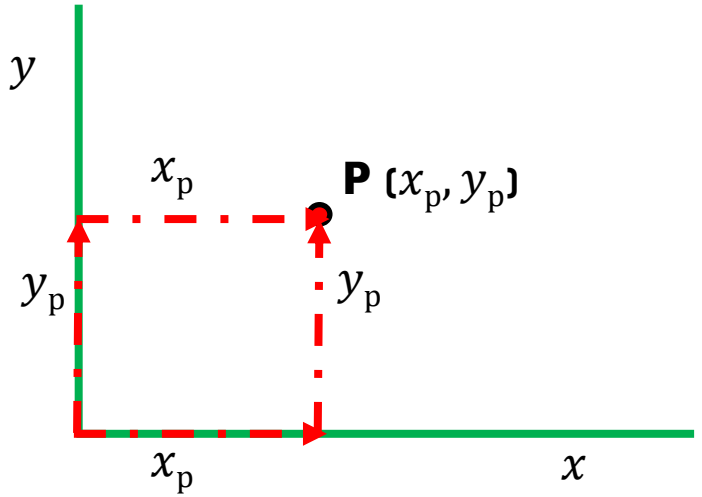
Cartesian Representation of Vectors – 2D

The Unit Vector

A unit vector is defined to be a dimensionless vector that has unit magnitude. Given an arbitrary vector $\vec{v}$ with non-zero magnitude $|\vec{v}|$, a unit vector may be constructed as:

$$\hat{u} = \frac{\vec{v}}{|\vec{v}|}$$

Cartesian coordinate system uses two orthogonal reference directions, usually denoted by *x* and y.



The intersection of the reference directions forms the origin of the coordinate system.

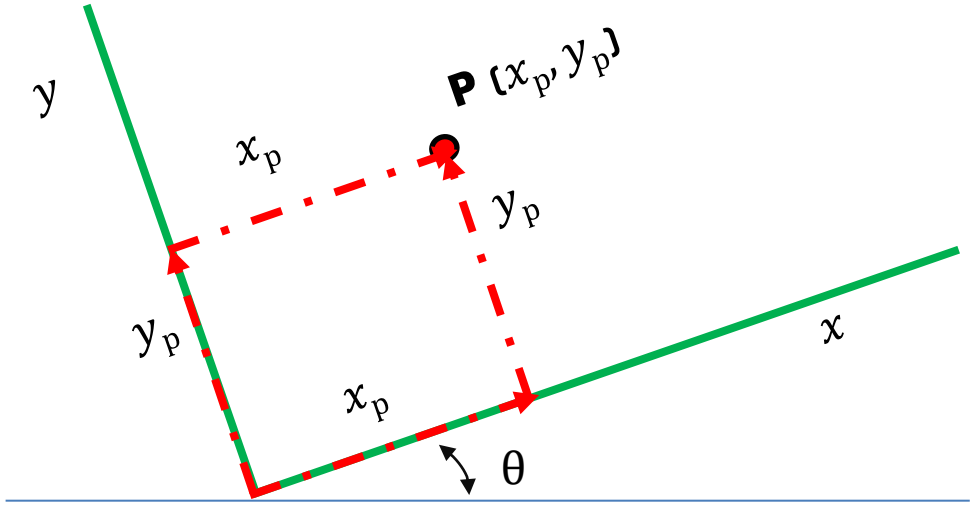
Coordinates of a point (P) are denoted by two ordered pair of numbers (x_p, y_p) where x_p and y_p are the x and y coordinates of the point respectively and measured from the origin.

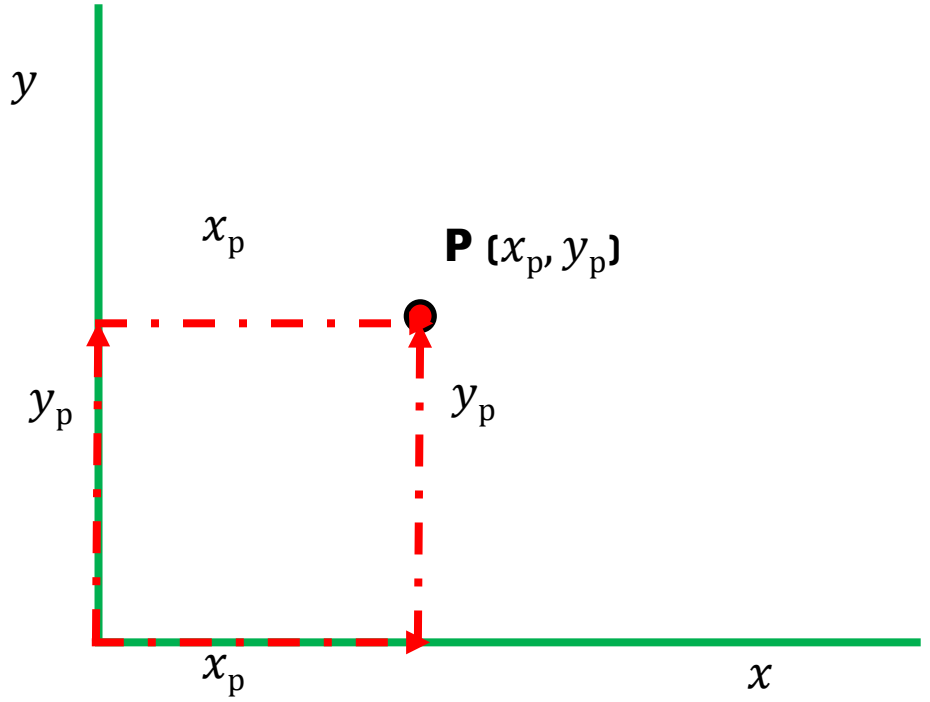
x and y directions need not be horizontal or vertical. The key requirement is orthogonality.

Unit Vector

x **and** ***y*** **directions need not be horizontal or vertical.**
The key requirement is orthogonality.

IMPLICATION

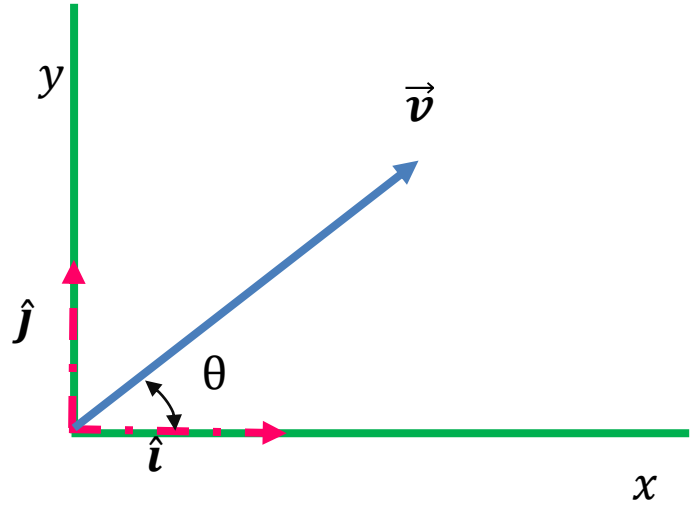




One could validly rotate axes through an angle θ with respect to the horizontal.

Unit Vector

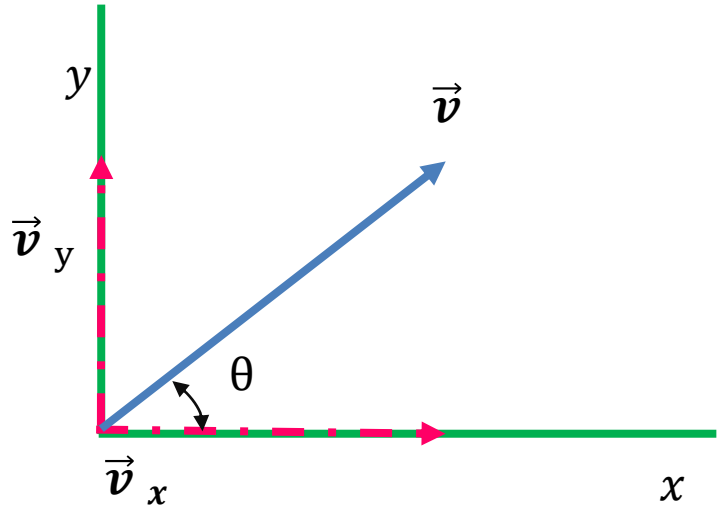
Unit vectors, $\hat{\imath}$ and $\hat{\jmath}$ are defined in the positive x and y respectively.



$\hat{\imath}$ and $\hat{\jmath}$ are both dimensionless and have directional significance (same way as east or north are used).

For a vector $\vec{v}$ in the xy plane, it may be written as:

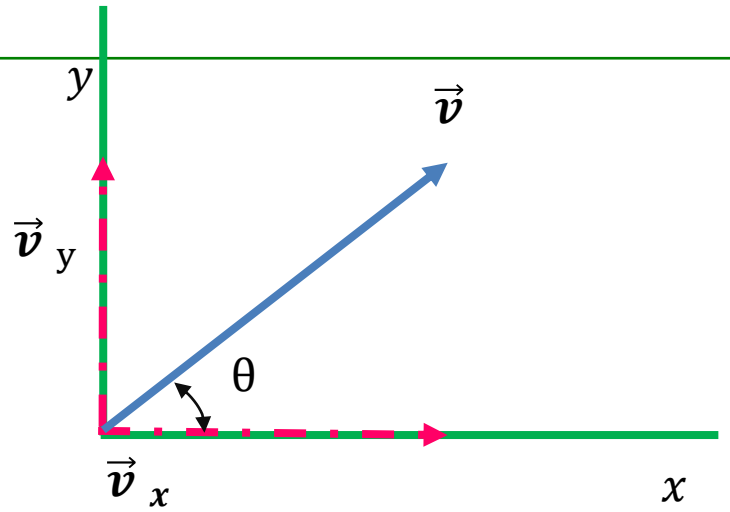
$$\vec{v} = \vec{v}_x + \vec{v}_y$$



Where $\vec{v}_x$ and $\vec{v}_y$ are vectors parallel to x and y directions respectively

Unit Vector

$$\vec{v} = \vec{v}_x + \vec{v}_y$$



$\vec{v}_x$ and $\vec{v}_y$ are referred to as the **vector components** of $\vec{v}$

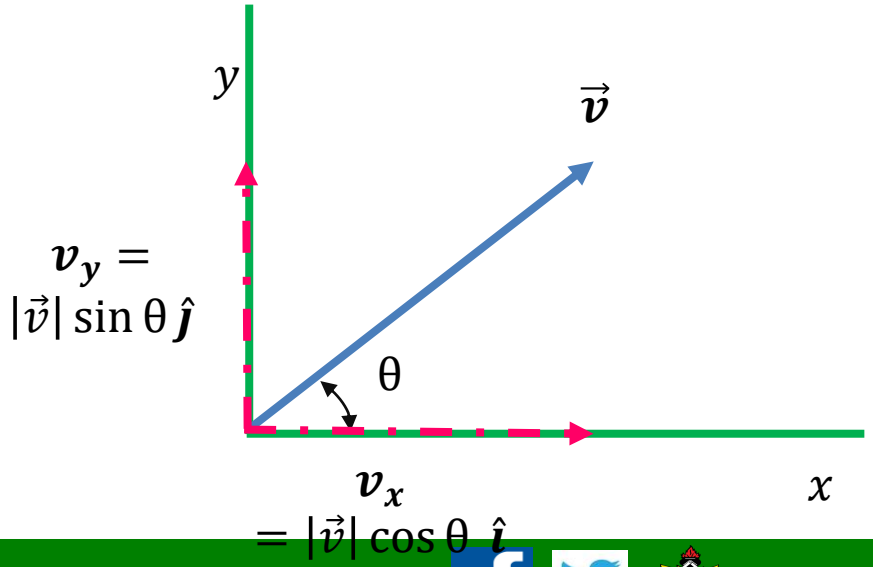
It may also be written as:

$$\vec{v} = v_x\hat{i} + v_y\hat{j}$$

Where, v_x and v_y are scalar components of $\vec{v}$ and given as:

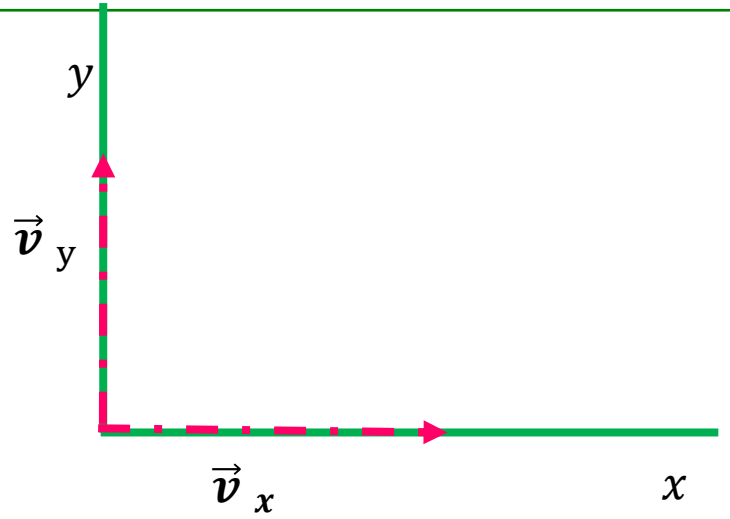
$$v_x = |\vec{v}|\cos\theta \qquad v_y = |\vec{v}|\sin\theta$$

$$\vec{v} = |\vec{v}|\cos\theta\,\hat{i} + |\vec{v}|\sin\theta\,\hat{j}$$



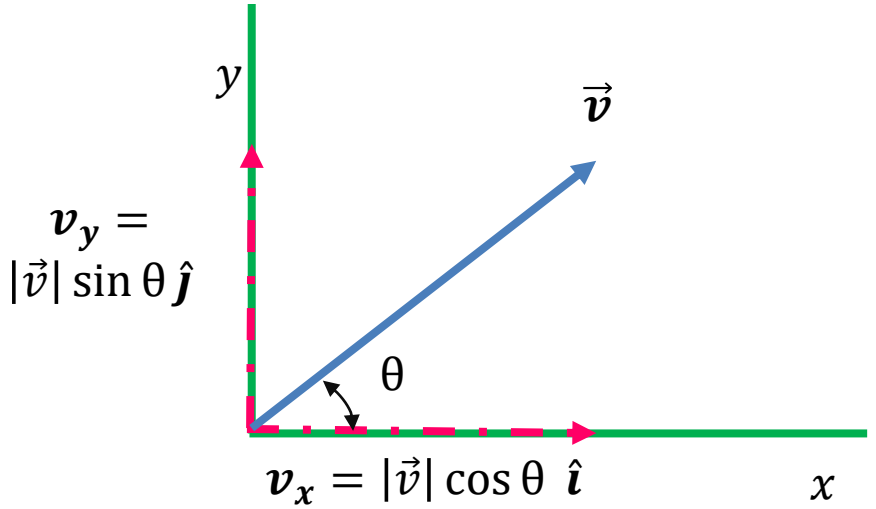
Unit Vector

Vector components are always orthogonal, therefore basic trigonometric relations and Pythagoras Theorem are applicable



The magnitude of vector $\vec{v}$ may be determined as:

$$|\vec{v}| = \sqrt{v_x^2 + v_y^2}$$



The angle with the horizontal becomes:

$$\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right)$$

Basic Vector Operations

$$\vec{A} = \left(6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k}\right) \text{m}$$

The **magnitude** (resultant) of the vector $\vec{A}$ may be determined as:

$$\left|\vec{A}\right| = \sqrt{6^2 + 3^2 + (-1)^2}$$

$$= 6.78 \text{ m}$$

A unit vector, $\hat{u}$ can also be defined on the basis of vector $\vec{A}$, as:

$$\hat{u} = \frac{\vec{A}}{\left|\vec{A}\right|}$$

$$= \frac{\left(6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k}\right) m}{\sqrt{(6\,m)^2 + (3\,m)^2 + (-1\,m)^2}}$$

$$\hat{u} = \frac{\left(6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k}\right)}{6.78}$$

Vector Addition – Cartesian Components

Given two vectors $\vec{v}_1$ and $\vec{v}_2$ such that:

$$\vec{v}_1 = v_{1x}\hat{\imath} + v_{1y}\hat{\jmath}$$

$$\vec{v}_2 = v_{2x}\hat{\imath} + v_{2y}\hat{\jmath}$$

The resultant $\vec{R}$, of these vectors is the sum of $\vec{v}_1$ and $\vec{v}_2$

$$\vec{R} = \vec{v}_1 + \vec{v}_2 \Rightarrow \vec{R} = \left(v_{1x}\hat{\imath} + v_{1y}\hat{\jmath}\right) + \left(v_{2x}\hat{\imath} + v_{2y}\hat{\jmath}\right)$$

$$\vec{R} = (v_{1x} + v_{2x})\hat{\imath} + (v_{1y} + v_{2y})\hat{\jmath} \Rightarrow \vec{R} = \vec{v}_1 + \vec{v}_2 + \cdots + \vec{v}_n$$

$$= \left(\sum_{i=1}^{n} v_{ix}\right)\hat{\imath} + \left(\sum_{i=1}^{n} v_{iy}\right)\hat{\jmath}$$

Generalized vector addition in 2-D

We shall examine applications of these in our next class.

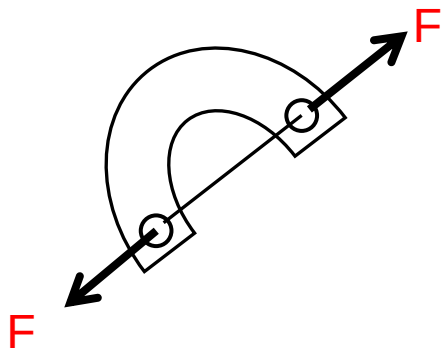
3D Force Systems

Force Systems

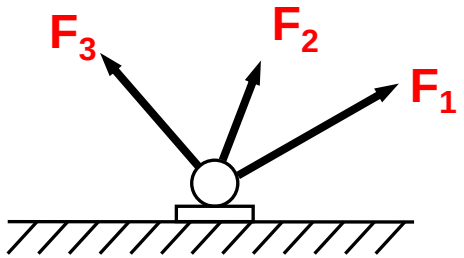
Spatial when all the forces do not lie in the same plane.

Forces may be:

concurrent if the lines of action of the forces intersect at a common point

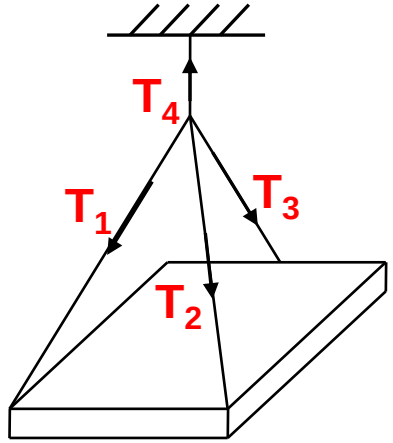


Collinear if the forces have a common line of action.

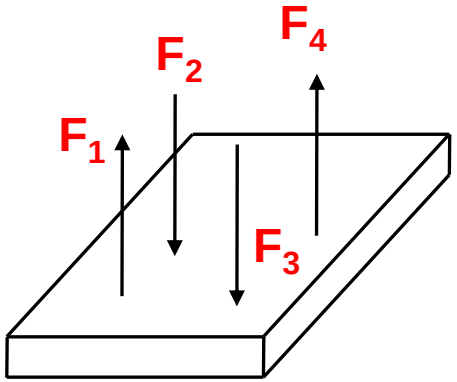


Concurrent Coplanar Force System

Coplanar when all the forces lie in the same plane.

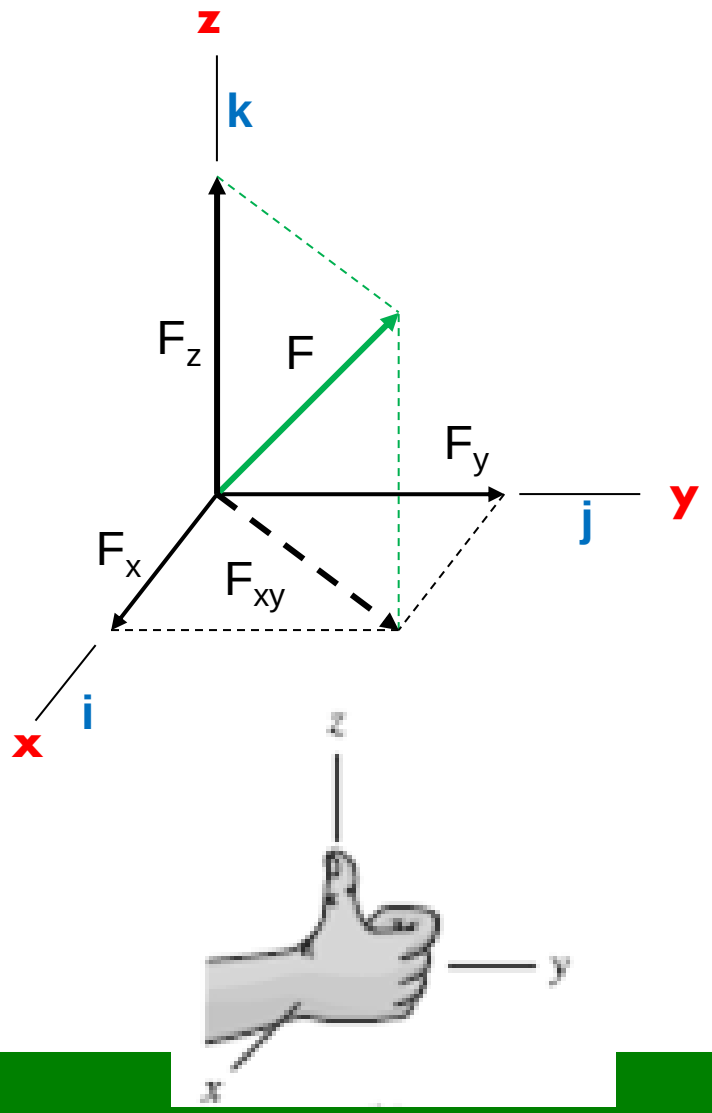


Concurrent Spatial Force System



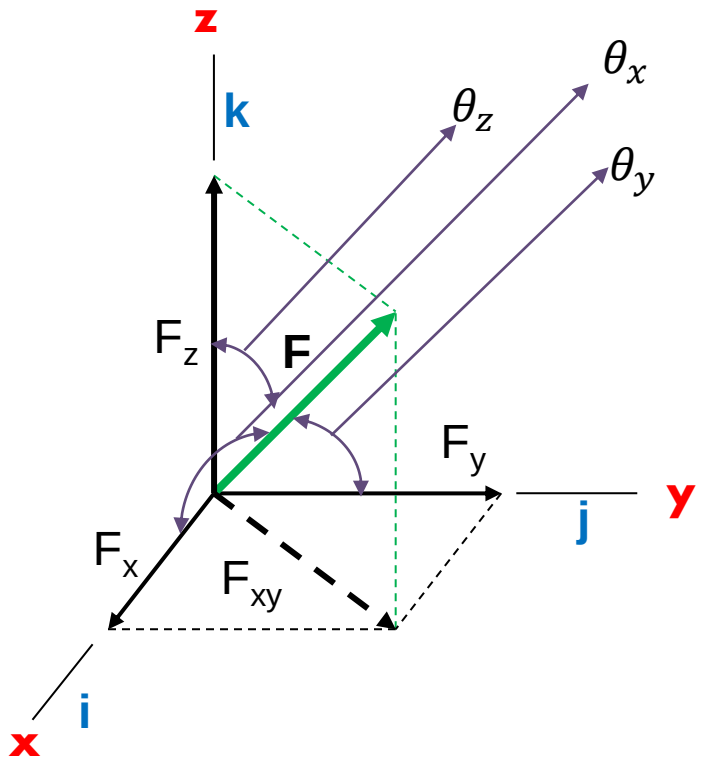
Parallel if the lines of action of the forces are parallel.

Cartesian Vector – 3D



- In right-handed Cartesian coordinate system
- The right thumb points in the positive **z** direction when the right hand figures are curled from positive **x** direction to positive **y** direction about the **z** axis.
- The three axes (**x**, **y** and **z**) are orthogonal (right angles) to each other.
- **i**, **j** and **k** are unit vectors along the axes **x**, **y** and **z**, respectively.

Cartesian Vector - 3D



$$\vec{F} = F_x i + F_y j + F_z k$$

θ_x, θ_y and θ_z are the angles between the resultant force and the positive x, y and z directions respectively

Then:

$$\cos\theta_x = \frac{F_x}{|\vec{F}|}$$

$$\cos\theta_y = \frac{F_y}{|\vec{F}|}$$

$$\cos\theta_z = \frac{F_z}{|\vec{F}|}$$

Direction cosines

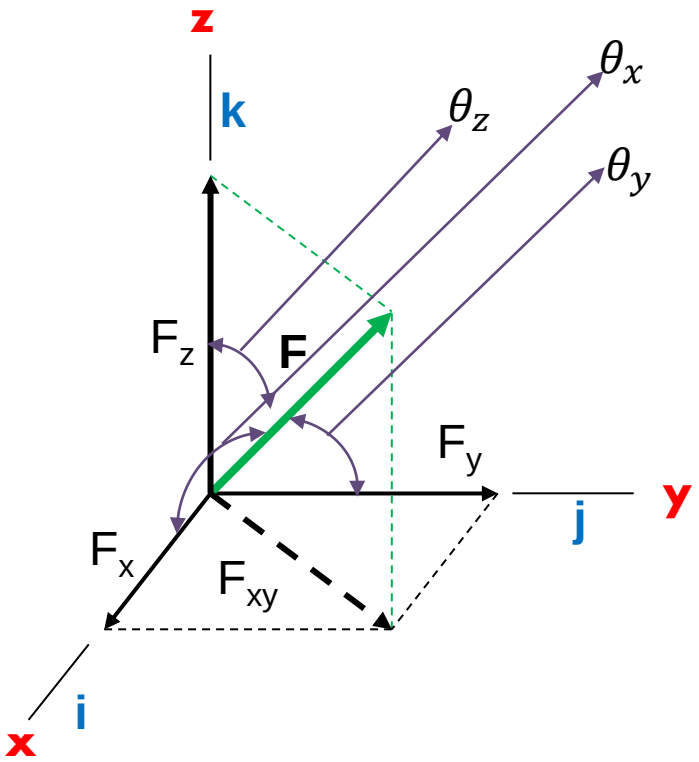
Where:

$$|\vec{F}| = \sqrt{F_x{}^2 + F_y{}^2 + F_z{}^2}$$

The unit vector may then be written as:

$$\vec{u} = \frac{F_x i + F_y j + F_z k}{|\vec{F}|}$$

Cartesian Vector - 3D



$$\vec{F} = F_x i + F_y j + F_z k$$

$$\vec{F} = |\vec{F}| cos\,\theta_x\, i + |\vec{F}| cos\,\theta_y\, j + |\vec{F}| cos\,\theta_z\, k$$

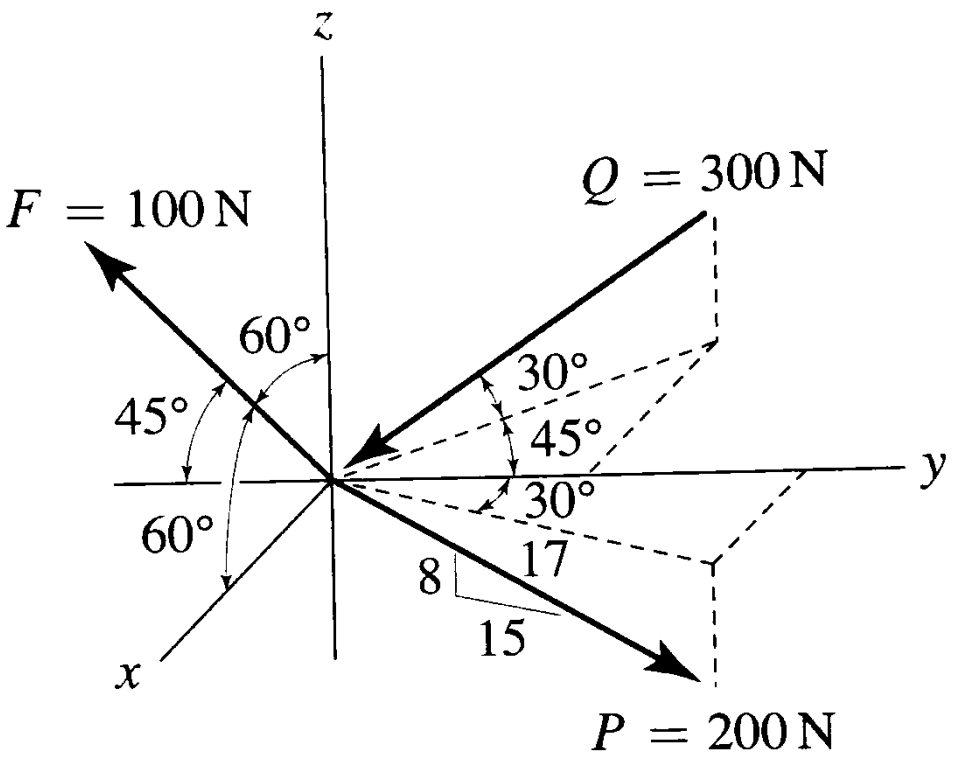
$$\vec{F} = |\vec{F}| \left(cos\,\theta_x\, i + cos\,\theta_y\, j + cos\,\theta_z\, k \right)$$

This is a unit vector pointing in the direction of $\vec{F}$

Also, it can be shown, that:

$$cos^2\theta_x + cos^2\theta_y + cos^2\theta_z = 1$$

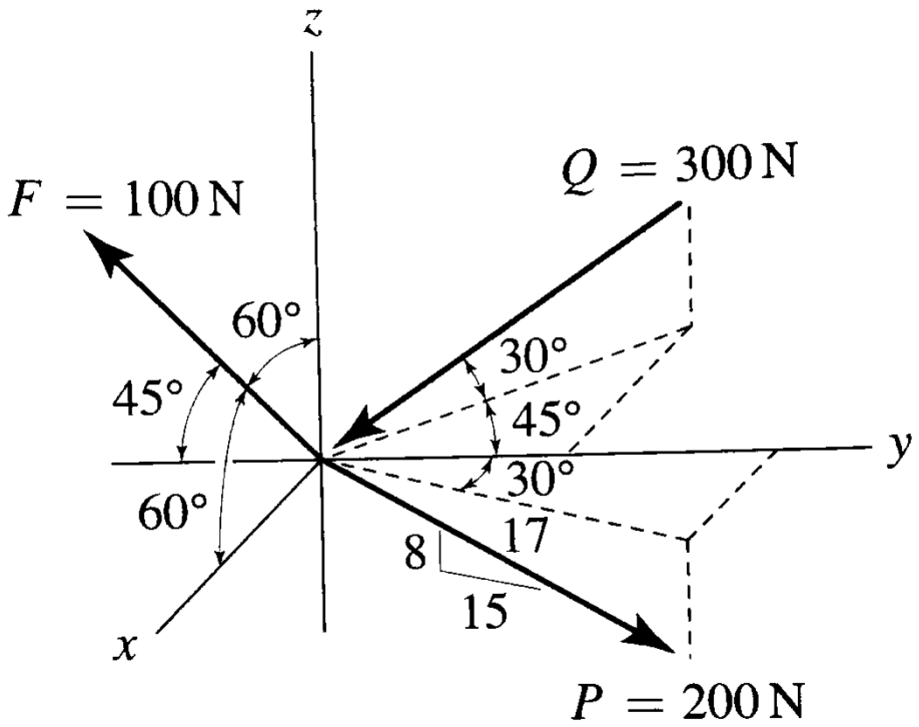
Example



Forces $\vec{F}$, $\vec{P}$ and $\vec{Q}$ (in Figure shown) are given a variety of different forms and orientation information that are commonly encountered.

1. **Write a vector expression for each force.**
2. **Determine the resultant of all three forces.**
3. **Determine the direction angles for the resultant.**

Example Force $\vec{F}$



We first look for direction angles:

$$\theta_x = 60^o$$

$$\theta_z = 60^o$$

$$\theta_y = 180^o - 45^o = 135^o$$

A useful check

$$\cos^2\theta_x + \cos^2\theta_y + \cos^2\theta_z = 1$$

$$\cos^2 60^o + \cos^2 135^o + \cos^2 60^o = 1$$

$$\vec{F} = 100\ N\left(\cos 60^o\ i + \cos 135^o\ j + \cos 60^0\ k\right)$$

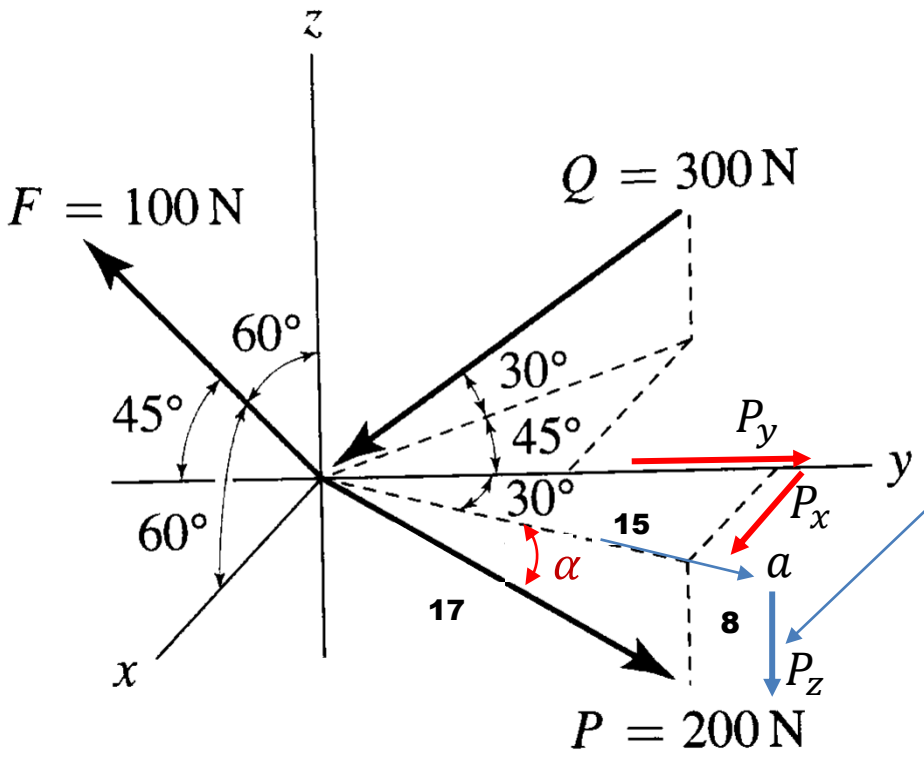
$$\vec{F} = \left(50.00i - 70.71j + 50.00k\right)N$$

Figure has been slightly modifed to clarify the position magnitudes presented as 15, 8 and 17.

Force $\vec{P}$

A 2-step process is adopted

Recognize first of all, that, 30° is not a directional angle.



The z-axis component of P (P_z) may be found as:

$$P_z = -200 \sin\alpha = -200\frac{8}{17} = -94.12 \text{ N}$$

Arbitrary axis *a* is defined, such that:

$$P_a = 200 \cos\alpha = 200\frac{15}{17} = 176.5\ N$$

P_a has components in both x and y directions. Using the 30° information provided:

$$P_x = P_a \sin 30^o = 176.5 \times 0.5 = 88.24 \text{ N}$$

and

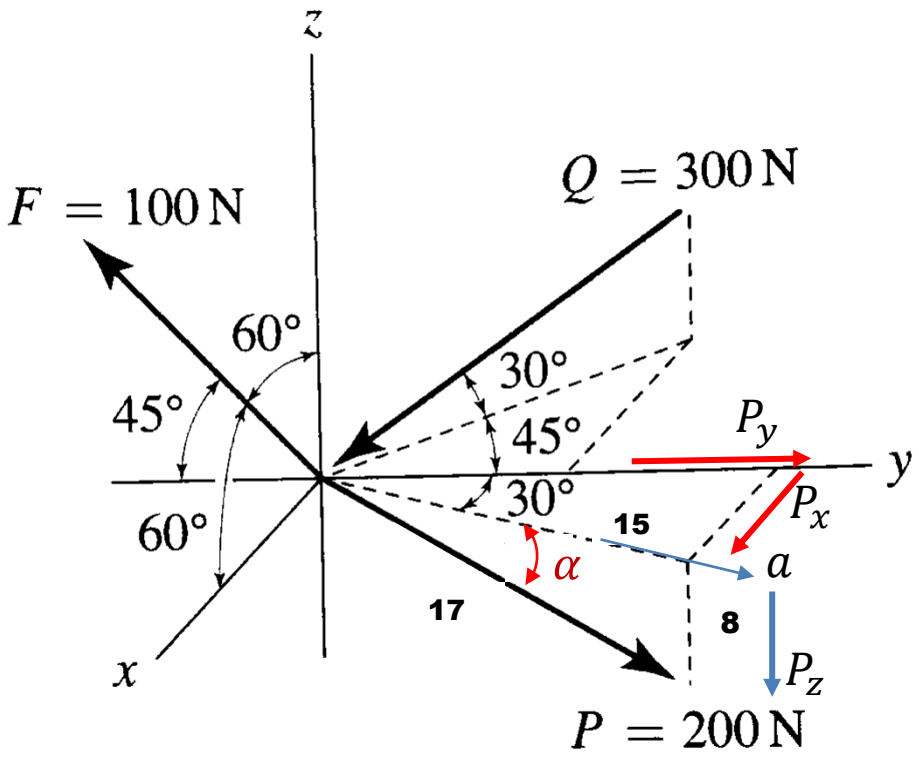
$$P_y = P_a \cos 30^o = 176.5 \times 0.866 = 152.8 \text{ N}$$

$$\vec{P} = P_x i + P_y j + P_z k$$

$$\vec{P} = (88.24i + 152.8j - 94.12k)\text{N}$$

Example

Force $\vec{P}$



$$\vec{\mathbf{P}} = 88.24\mathbf{i} + 152.8\mathbf{j} - 94.12\mathbf{k}$$

The result obtained may be verified by computing:

$$\mathbf{P} = \sqrt{\mathbf{P_x}^2 + \mathbf{P_y}^2 + \mathbf{P_z}^2}$$

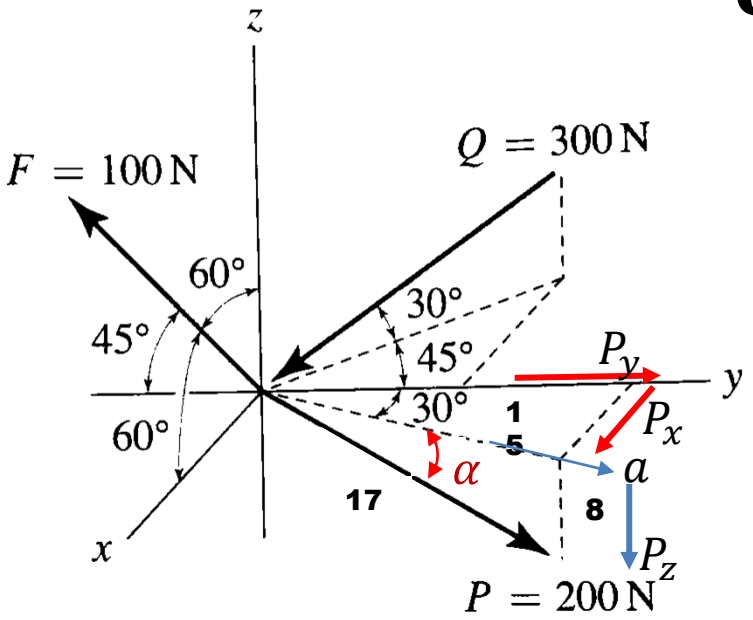
$$P = \sqrt{88.24^2 + 152.8^2 + (-94.12)^2}$$

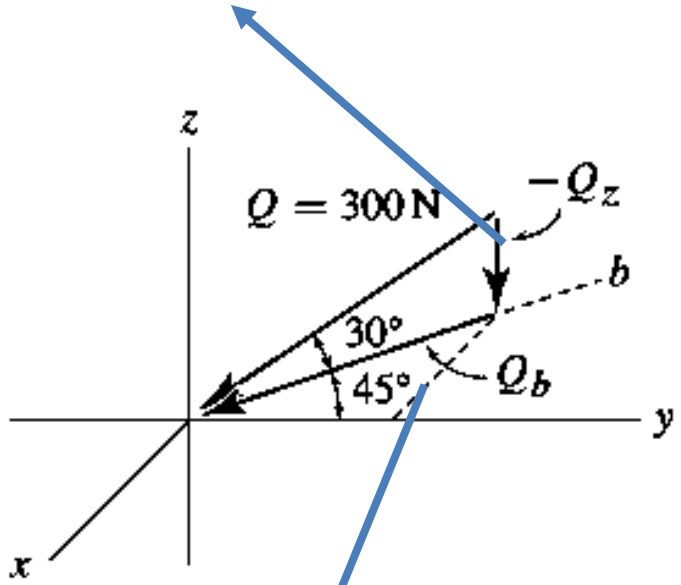
$$= 200\ \mathrm{N}$$

Force $\vec{Q}$ A similar 2-step process is adopted

The component of $\vec{Q}$ in the z direction is found as:

$$Q_z = -Q\ sin30o = -300(0.5) = -150\ \text{N}$$



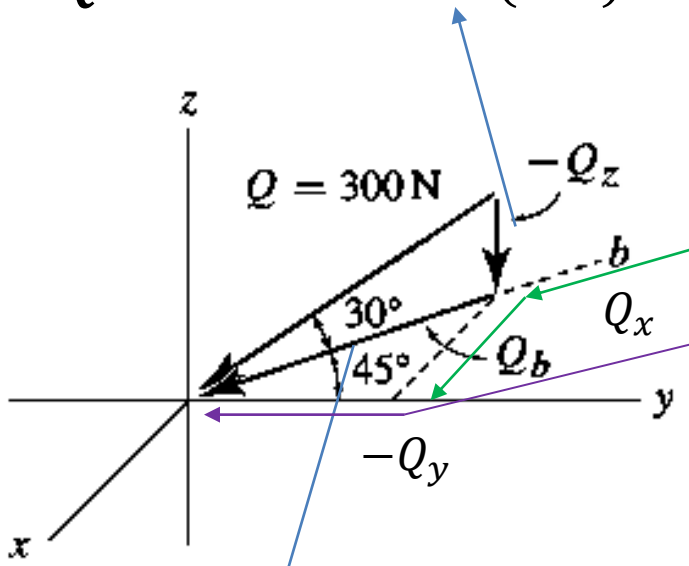


$$Q_b = -Q\ cos30o$$
$$= -300(0.866) = -259.8\ \text{N}$$

Force $\vec{Q}$

A similar 2-step process is adopted with Q_b, and using the 45° information provided, x component is found as:

$$Q_z = -Q\,sin30o = -300(0.5) = -150\ \text{N}$$



$$Q_b = -Q\,cos30o = -300(0.866) = -259.8\ \text{N}$$

$$Q_x = Q_b(sin45^o) = 259.8(0.707) = 183.7\ \text{N}$$

$$Q_y = Q_b cos45o = -259.8(0.707) = -183.7\ \text{N}$$

$$\vec{Q} = 183.7\text{i} - 183.7\text{j} - 150.0\text{k}$$

Once again a check may be run by verifying that:

$$Q = \sqrt{Q_x^{\,2} + Q_y^{\,2} + Q_z^{\,2}}$$

Resultant force $\vec{R}$

$$\vec{R} = \vec{F} + \vec{P} + \vec{Q}$$

$(50.00\mathbf{i} - 70.71\mathbf{j} + 50.00\mathbf{k})$

$(88.24\mathbf{i} + 152.8\mathbf{j} - 94.12\mathbf{k})$

$(183.7\mathbf{i} - 183.7\mathbf{j} - 150.0\mathbf{k})$

$$\vec{\mathbf{R}} = (50.00\mathbf{i} - 70.71\mathbf{j} + 50.00\mathbf{k}) + (88.24\mathbf{i} + 152.8\mathbf{j} - 94.12\mathbf{k}) + (183.7\mathbf{i} - 183.7\mathbf{j} - 150.0\mathbf{k})$$

$$\vec{\mathbf{R}} = (50.00 + 88.24 + 183.7)\mathbf{i} + (-70.71 + 152.8 - 183.7)\mathbf{j} + (50.00 - 94.12 - 150.00)\mathbf{k}$$

$$\vec{\mathbf{R}} = (321.9\,\mathbf{i} - 101.6\,\mathbf{j} - 194.1\,\mathbf{k})$$

$$|\vec{R}| = \sqrt{321.9^2 + (-101.6)^2 + (-194.1)^2} = 389.4\ \mathrm{N}$$

Orientation of Resultant force $\vec{R}$:

$$\vec{R} = (321.9\,i - 101.6\,j - 194.1\,k)\ N$$

$$\cos\theta_x = \frac{R_x}{|\vec{R}|} = \frac{321.9}{389.4} \Rightarrow \theta_x = \cos^{-1}\left(\frac{321.9}{389.4}\right) = 34.2^o$$

$$\cos\theta_y = \frac{R_y}{|\vec{R}|} = \frac{-101.6}{389.4} \Rightarrow \theta_y = \cos^{-1}\left(\frac{-101.6}{389.4}\right) = 105.1^o$$

$$\cos\theta_z = \frac{R_z}{|\vec{R}|} = \frac{-194.1}{389.4} \Rightarrow \theta_z = \cos^{-1}\left(\frac{-194.1}{389.4}\right) = 119.9^o$$